Ex no: 3	Continuous Integration/Continuous Deployment of a React Application using GitHub and Jenkins Freestyle Project
Date:	

Aim:

To automate the build and deployment process of a React-based **Student Management System** using **GitHub** for version control and **Jenkins Freestyle Project** for Continuous Integration/Continuous Deployment (CI/CD).

Procedure :

Step 1: Create GitHub Repository and Clone

1. Create a new repository in GitHub named:

Student-Management-System

2. Clone the repository into the local system inside a folder such as devops using:

git clone https://github.com/antonyrajinevesa-del/Student-Management-System.git

3. Navigate into the cloned repository:

cd Student-Management-System

4. Open the project folder in **Visual Studio Code**.

Step 2: Create a New React Project

1. Open the terminal in Visual Studio Code inside the cloned repository folder.
2. Create a React project using Vite:

npm create vite@latest .

3. Select the required options:

Framework: React
Variant: JavaScript

4. Install the required project dependencies:
npm install
5. Run the application locally to verify the React application:

npm run dev

6. Open the application in the browser using the displayed local URL:

http://localhost:5173/

7. Stop the development server using:

Ctrl + C

Step 3: Build the React Application

1. Generate the production-ready build using:

```
npm run build
```

2. The Vite build process generates the production files inside the dist folder.
3. The generated structure contains files similar to:

```
dist/
├── index.html
└── assets/
    ├── index-xxxxx.css
    └── index-xxxxx.js
```

4. The production build was successfully generated with Vite.

Step 4: Push the Project to GitHub

1. Add the project files to Git:

```
git add .
```

2. Commit the project:

```
git commit -m "feat: student management system added"
```

3. Push the project to the main branch:

```
git push origin main
```

4. Verify that the Student Management System source code is available in the GitHub repository.

Step 5: Configure Jenkins Job

1. Open Jenkins in the browser:

http://localhost:8080

2. Select:

New Item → Freestyle Project

3. Enter the project name:

```
StudentManagementSystem
```

4. In **Source Code Management**, select:

```
Git
```

5. Enter the GitHub repository URL:

https://github.com/antonyrajinevesa-del/Student-Management-System

6. Set the Branch Specifier as:

```
*/main
```

Jenkins clones the repository into its workspace before executing the build process.

Step 6: Add Build Steps

Since Jenkins is running on macOS, select:

Build Steps → Add build step → Execute shell

Add the following commands:

```
export PATH="/usr/local/bin:/usr/bin:/bin:/usr/sbin:/sbin:$PATH"

echo "Node location:"
which node

echo "npm location:"
which npm

echo "Node version:"
node --version

echo "npm version:"
npm --version

echo "Installing dependencies..."
npm ci || npm install

echo "Building React application..."
npm run build

echo "Deploying application..."
mkdir -p "$JENKINS_HOME/userContent/react-app"

rm -rf "$JENKINS_HOME/userContent/react-app/"*

cp -R dist/. "$JENKINS_HOME/userContent/react-app/"

echo "Deployment completed."
```

Where:

- export PATH=... → Makes Node.js and npm available to the Jenkins shell environment.
- which node → Displays the location of Node.js.
- which npm → Displays the location of npm.
- node --version → Displays the installed Node.js version.
- npm --version → Displays the installed npm version.
- npm ci || npm install → Installs project dependencies, using npm ci first and npm install as a fallback.
- npm run build → Creates the production-ready React application inside the dist folder.

- mkdir -p ... → Creates the Jenkins deployment directory.
- rm -rf ... → Removes the previous deployment files.
- cp -R dist/. ... → Copies the latest production build into Jenkins for deployment.

The reference experiment performs the same dependency installation, production build, and copying of dist into Jenkins' userContent directory, with Windows-specific commands.

Step 7: Build and Deploy

1. Save the Jenkins job.
2. Click:

Build Now

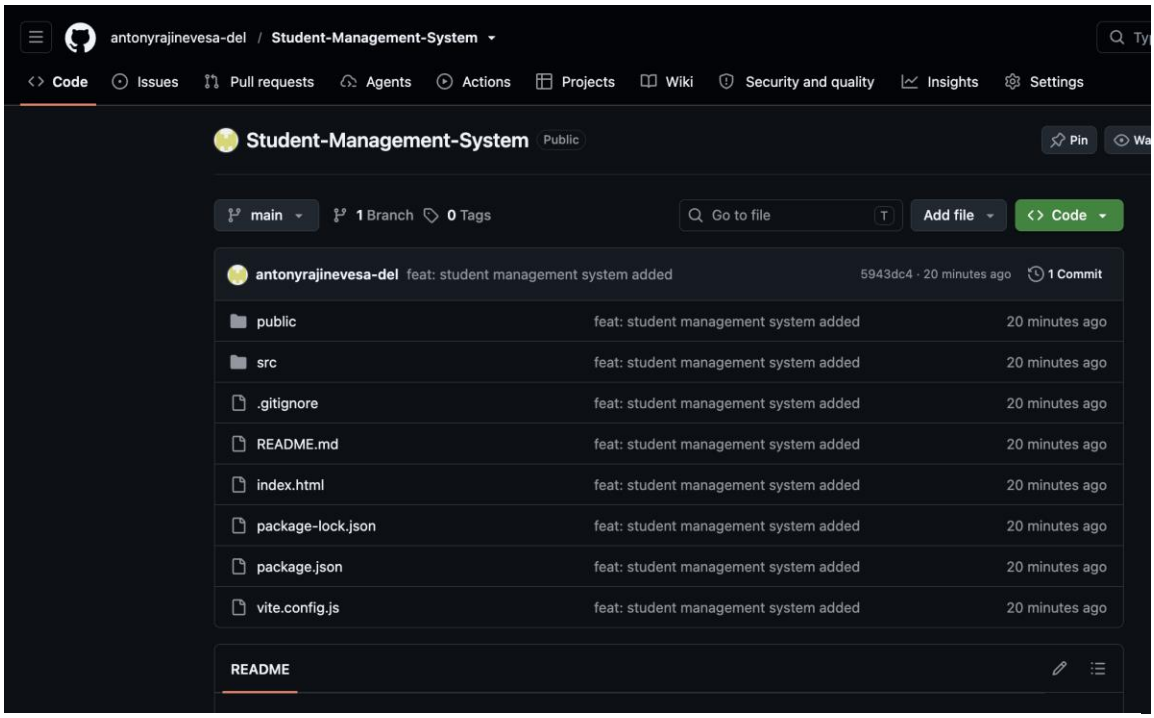
3. Jenkins clones the latest version of the project from GitHub.
4. Jenkins installs the required dependencies using npm.
5. Jenkins executes the production build using Vite.
6. The generated dist folder is copied into the Jenkins userContent deployment directory.
7. After a successful build, the Jenkins console displays:

Finished: SUCCESS

8. The deployed React application can be accessed through Jenkins using:

   http://localhost:8080/userContent/react-app/

OUTPUT :



Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell

Command

See the list of available environment variables

```
export PATH="/usr/local/bin:/usr/bin:/bin:/usr/sbin:/sbin:$PATH"

echo "Node location:"
which node

echo "npm location:"
which npm

echo "Node version:"
node --version

echo "npm version:"
npm --version

echo "Installing dependencies..."
npm ci || npm install

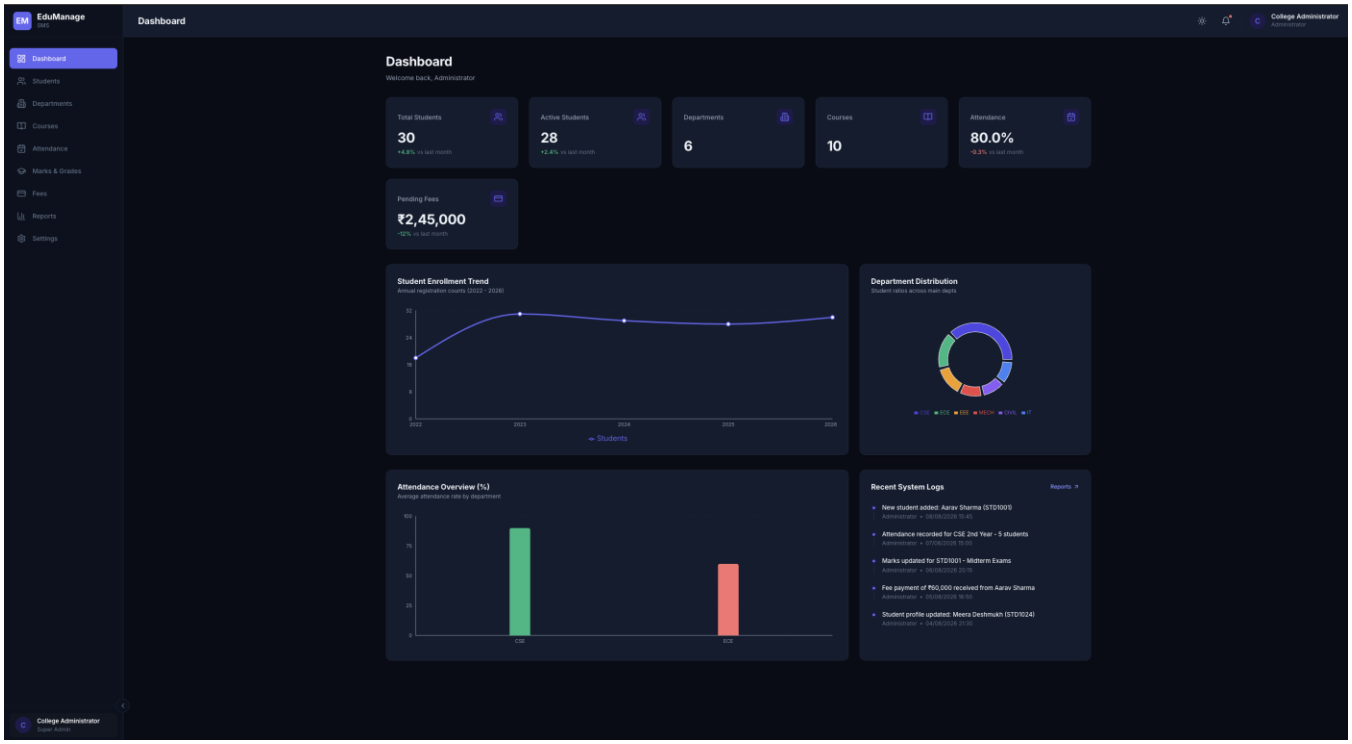
echo "Building React application..."
npm run build

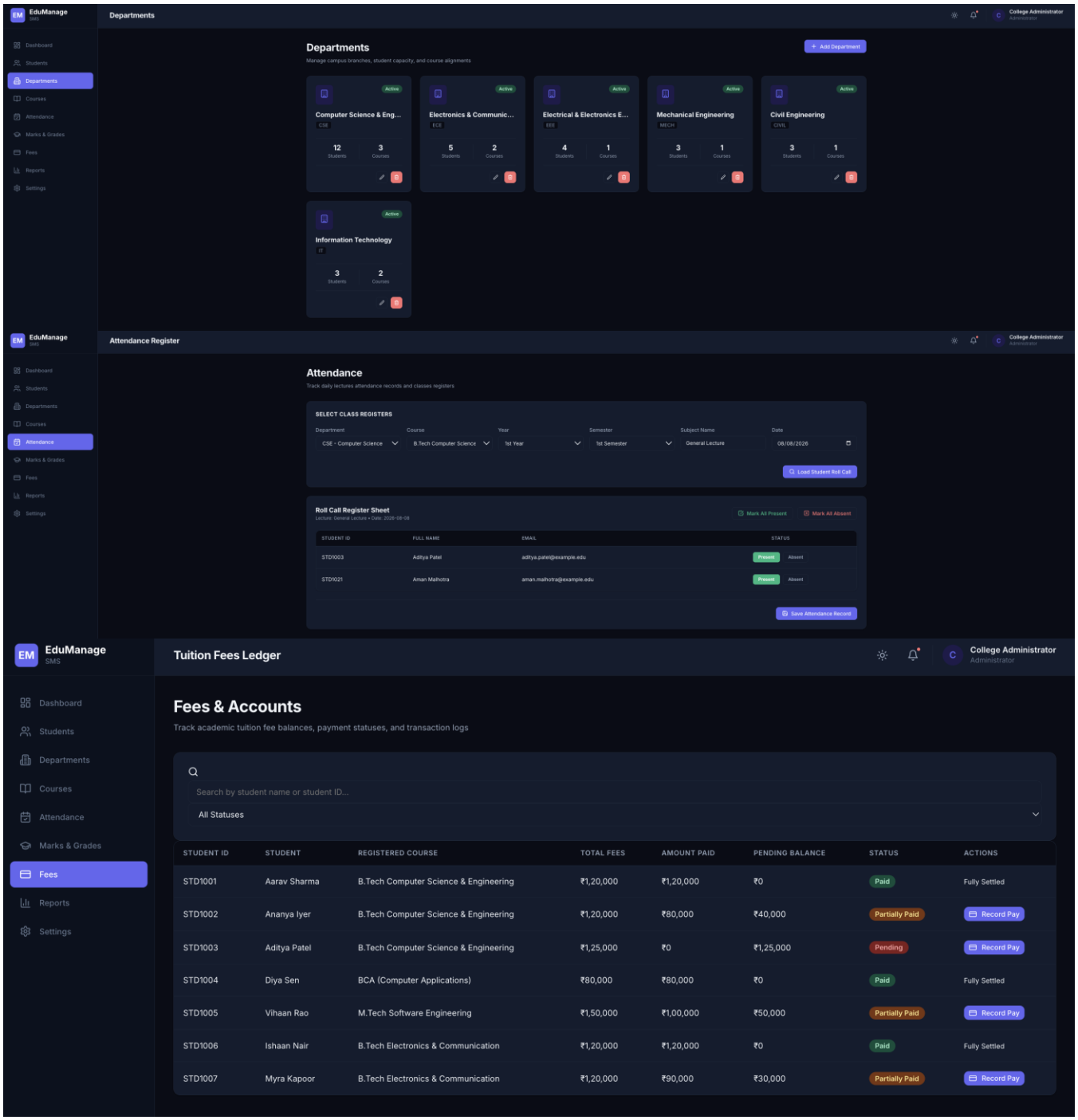
echo "Deploying application..."
mkdir -p "$JENKINS_HOME/userContent/react-app"

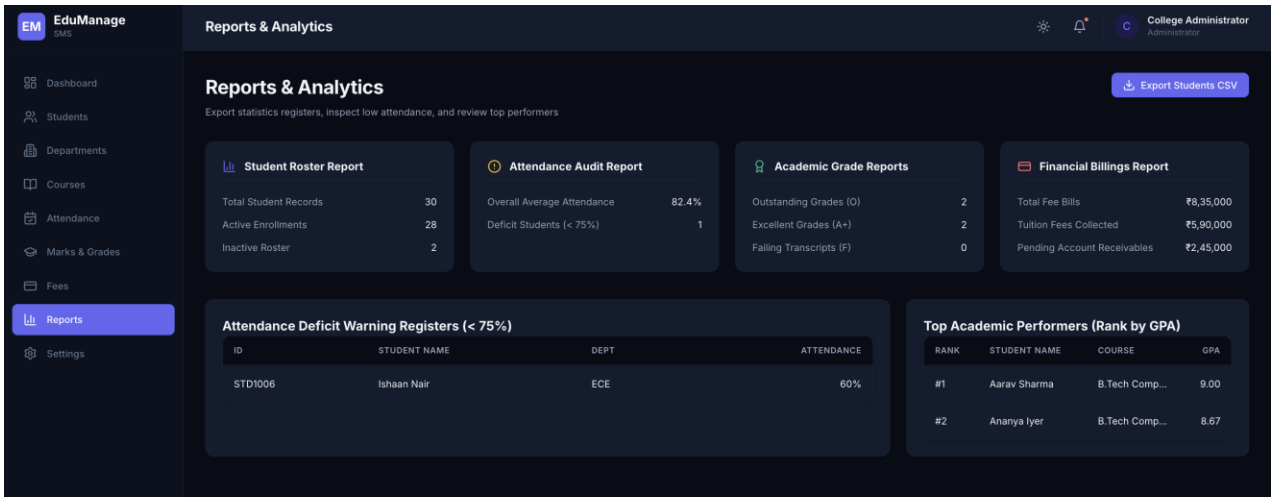
rm -rf "$JENKINS_HOME/userContent/react-app/"*

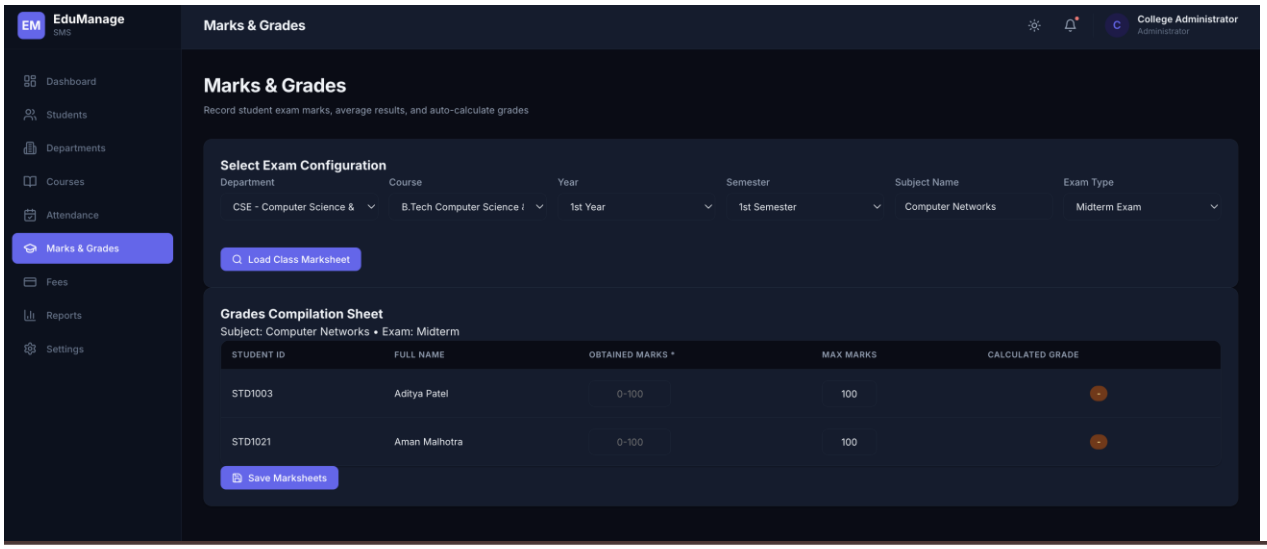
cp -R dist/. "$JENKINS_HOME/userContent/react-app/"

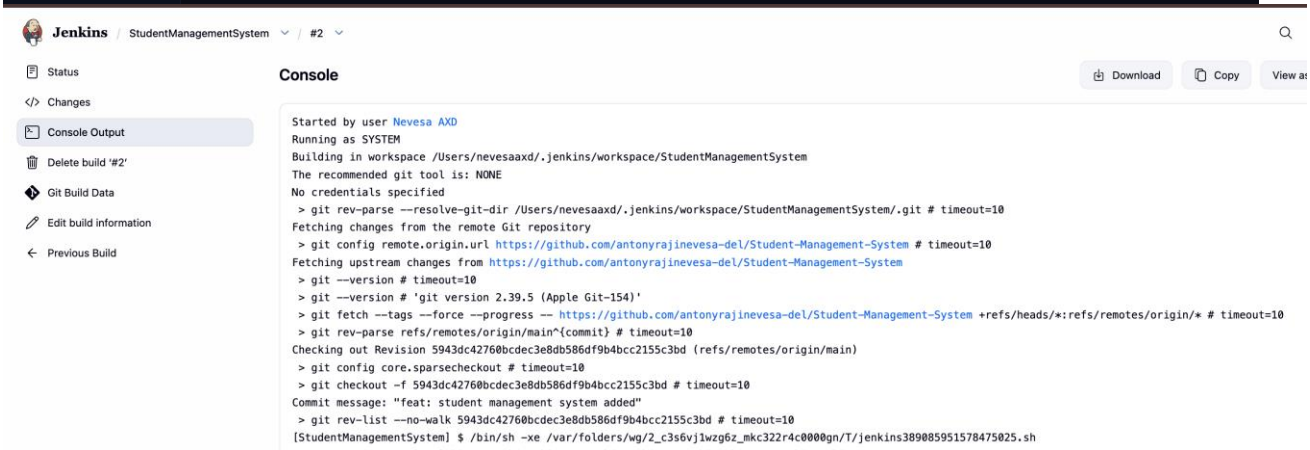
echo "Deployment completed."
```


EduManage
SMS
Dashboard
College Administrator
Administrator
Dashboard
Students
Departments
Courses
Attendance
Marks & Grades
Fees
Reports
Settings
College Administrator
Super Admin
Dashboard
Welcome back, Administrator
Total Students
30
+4.8% vs last month
Active Students
28
+2.4% vs last month
Departments
6
Courses
10
Attendance
80.0%
-0.3% vs last month
Pending Fees
₹2,45,000
-12% vs last month
Student Enrollment Trend
Annual registration counts (2022 – 2026)
32
24
16
8
0
2022
2023
2024
2025
2026
Students
Department Distribution
Student ratios across main depts
CSE
ECE
EEE
MECH
CIVIL
IT
Attendance Overview (%)
Average attendance rate by department
100
75
50
25
0
CSE
ECE
Recent System Logs
Reports
New student added: Aarav Sharma (STD1001)
Administrator • 08/08/2026 15:45
Attendance recorded for CSE 2nd Year - 5 students
Administrator • 07/08/2026 15:00
Marks updated for STD1001 - Midterm Exams
Administrator • 06/08/2026 20:15
Fee payment of ₹60,000 received from Aarav Sharma
Administrator • 05/08/2026 18:30
Student profile updated: Meera Deshmukh (STD1024)
Administrator • 04/08/2026 21:30


EduManage
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Departments
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Marks & Grades
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Settings
College Administrator
Administrator
Departments
Manage campus branches, student capacity, and course alignments
Add Department
Computer Science & Eng...
CSE
12
Students
3
Courses
Active
Electronics & Communic...
ECE
5
2
Electrical & Electronics E...
EEE
4
1
Mechanical Engineering
MECH
3
1
Civil Engineering
CIVIL
3
1
Information Technology
IT
3
2
Attendance Register
Attendance
Track daily lectures attendance records and classes registers
SELECT CLASS REGISTERS
Department
Course
Year
Semester
Subject Name
Date
CSE - Computer Science
B.Tech Computer Science
1st Year
1st Semester
General Lecture
08/08/2026
Load Student Roll Call
Roll Call Register Sheet
Mark All Present
Mark All Absent
STUDENT ID
FULL NAME
EMAIL
STATUS
STD1003
Aditya Patel
aditya.patel@example.edu
Present
Absent
STD1021
Aman Malhotra
aman.malhotra@example.edu
Present
Absent
Save Attendance Record
Tuition Fees Ledger
Fees & Accounts
Track academic tuition fee balances, payment statuses, and transaction logs
Search by student name or student ID...
All Statuses
STUDENT ID
STUDENT
REGISTERED COURSE
TOTAL FEES
AMOUNT PAID
PENDING BALANCE
STATUS
ACTIONS
STD1001
Aarav Sharma
B.Tech Computer Science & Engineering
₹1,20,000
₹1,20,000
₹0
Paid
Fully Settled
STD1002
Ananya Iyer
B.Tech Computer Science & Engineering
₹1,20,000
₹80,000
₹40,000
Partially Paid
Record Pay
STD1003
Aditya Patel
B.Tech Computer Science & Engineering
₹1,25,000
₹0
₹1,25,000
Pending
Record Pay
STD1004
Diya Sen
BCA (Computer Applications)
₹80,000
₹80,000
₹0
Paid
Fully Settled
STD1005
Vihaan Rao
M.Tech Software Engineering
₹1,50,000
₹1,00,000
₹50,000
Partially Paid
Record Pay
STD1006
Ishaan Nair
B.Tech Electronics & Communication
₹1,20,000
₹1,20,000
₹0
Paid
Fully Settled
STD1007
Myra Kapoor
B.Tech Electronics & Communication
₹1,20,000
₹90,000
₹30,000
Partially Paid
Record Pay


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Reports & Analytics
Export statistics registers, inspect low attendance, and review top performers
Export Students CSV
Student Roster Report
Total Student Records 30
Active Enrollments 28
Inactive Roster 2
Attendance Audit Report
Overall Average Attendance 82.4%
Deficit Students (< 75%) 1
Academic Grade Reports
Outstanding Grades (O) 2
Excellent Grades (A+) 2
Failing Transcripts (F) 0
Financial Billings Report
Total Fee Bills ₹8,35,000
Tuition Fees Collected ₹5,90,000
Pending Account Receivables ₹2,45,000
Attendance Deficit Warning Registers (< 75%)
ID STUDENT NAME DEPT ATTENDANCE
STD1006 Ishaan Nair ECE 60%
Top Academic Performers (Rank by GPA)
RANK STUDENT NAME COURSE GPA
#1 Aarav Sharma B.Tech Comp... 9.00
#2 Ananya Iyer B.Tech Comp... 8.67


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Reports
Settings
Marks & Grades
Record student exam marks, average results, and auto-calculate grades
Select Exam Configuration
Department
CSE - Computer Science &
Course
B.Tech Computer Science
Year
1st Year
Semester
1st Semester
Subject Name
Computer Networks
Exam Type
Midterm Exam
Load Class Marksheet
Grades Compilation Sheet
Subject: Computer Networks • Exam: Midterm
STUDENT ID FULL NAME OBTAINED MARKS * MAX MARKS CALCULATED GRADE
STD1003 Aditya Patel 0-100 100 -
STD1021 Aman Malhotra 0-100 100 -
Save Marksheets


Jenkins
StudentManagementSystem
#2
Status
Changes
Console Output
Delete build '#2'
Git Build Data
Edit build information
Previous Build
Console
Download
Copy
View as
Started by user Nevesa AXD
Running as SYSTEM
Building in workspace /Users/nevesaaxd/.jenkins/workspace/StudentManagementSystem
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /Users/nevesaaxd/.jenkins/workspace/StudentManagementSystem/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/antonyrajinevesa-del/Student-Management-System # timeout=10
Fetching upstream changes from https://github.com/antonyrajinevesa-del/Student-Management-System
> git --version # timeout=10
> git --version # 'git version 2.39.5 (Apple Git-154)'
> git fetch --tags --force --progress -- https://github.com/antonyrajinevesa-del/Student-Management-System +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 5943dc42760bcdec3e8db586df9b4bcc2155c3bd (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 5943dc42760bcdec3e8db586df9b4bcc2155c3bd # timeout=10
Commit message: "feat: student management system added"
> git rev-list --no-walk 5943dc42760bcdec3e8db586df9b4bcc2155c3bd # timeout=10
[StudentManagementSystem] $ /bin/sh -xe /var/folders/wg/2_c3s6vj1wzg6z_mkc322r4c0000gn/T/jenkins38908595157847502 5.sh

GitHub commands syntax & Description (10)	Implementation and Execution (15)	Tool/ Technology usage (10)	Code Quality & Documentation (5)	Time Management (5)	Viva (5)	Total (50)

Result:

The React application was successfully integrated with GitHub and Jenkins Freestyle Project. The build and deployment process was automated, and the application was successfully deployed and executed locally.